

5	C	By the Principle of Floatation (or by considering the net force on the cube), the upthrust on the cube is equal to the weight of the cube. weight of cube = $mg = \rho_0 Vg = \rho_0 L^3 g$
6	D	The shelf is light, so its weight can be neglected. The centre of gravity of the system is thus at the